

TUTORLINKAI: A Verified Local Tutor Discovery and Booking Platform

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Abstract- *In the fast-paced world, education plays a vital role in shaping the future of the younger generation. Many students find it difficult to keep up with their classes and compete academically with their peers. This creates a need for additional academic support for the students, leading to parents seeking tutors who can help their children understand the concepts where they require more assistance.*

Parents often choose tutors who are close to their homes or tutors who have been suggested by relatives or neighbours; they miss out on checking if the tutor is the right fit for their child.

It is given that the parents want what is best for their child and help them academically, but they do not always perform background checks and assess whether this tutor is appropriate for the needs of their child and if they are properly qualified.

This often leads to the child not being able to understand the concepts despite the parents providing them with some help, but just not the right resources.

Another major problem that parents and students face is the lack of verification for the tutors. We must understand who is helping our children and assess if that person is legitimate and trustworthy. It is also important that we perform a credibility check rather than blindly believing the neighbors' or relatives' suggestions.

This is where TutorLinkAI comes in; the main goal of this application is to provide parents with the right information

to help provide the children with the right resources that help them academically without adding any strain.

TutorLinkAI focuses on connecting parents with the right tutors by assessing their qualifications and seeing if they can help in subjects where the child needs help.

Using TutorLinkAI, parents can check if the tutor is actually credible and has the appropriate degrees and credentials to help their children; this creates a sense of relief for both the children and the parents.

Keywords: TutorLinkAI, Online Tutoring, Tutor Verification, Credential Verification, Optical Character Recognition (OCR), Artificial Intelligence, Tutor Recommendation, Educational Technology, Academic Support, Trust and Safety

I. INTRODUCTION

Learning has increasingly moved beyond the confines of a traditional classroom. Given the academic background of students and their level of comprehension, classroom teaching may not always be enough and as such tuition services have long provided additional support to students who require more help in certain areas of learning than what is provided in the classroom. However, the efficiency

of such assistance depends on locating a tutor who is qualified, experienced, and able to provide guidance to the student according to their specific needs.

Traditionally, students and parents have relied on a number of private networks, including family, friends, neighbours in order to locate tutors. While such methods can be effective in shortlisting potential tutors, the information provided is usually limited. Factors to consider when selecting a tutor include qualifications, the subject matter or area of expertise that the student needs help with, experience, location, availability, and fees amongst other considerations. Gathering and comparing this information can be a tedious process which many parents and students find inconvenient.

Online services have been developed to enable users to first research and then locate a suitable service based on certain criteria. However, there are various limitations associated with existing tutor discovery methods and services. Firstly, many existing platforms do not allow parents or students to easily locate tutors using certain criteria, as well as comparing tutors based on certain requirements. Such an option can prove especially useful to parents and students looking for specific tutors. In many cases, finding a tutor relies heavily on word of mouth, which limits students and parents to tutors within their personal networks and provides limited opportunities to match tutors effectively with the specific needs and requirements of the student.

Even further into the tutor discovery process, parents and students must engage in separate communication channels when contacting a tutor. As such, parents and students must individually contact a tutor in order to confirm their qualifications, availability, as well as schedule and reschedule sessions. This can be inconvenient and in many cases redundant. A proper platform can enable both students and parents to communicate with tutors and manage scheduling and other activities related to the tutoring sessions.

An equally important limitation in the tutor discovery process is the reliability of information provided to parents and students. It is vital that both students and parents consider a number of factors when selecting a tutor, including educational background, experience, and the area of expertise. Making such information easily accessible to parents and students can be extremely valuable in ensuring that both parties have realistic expectations of the tutor's qualifications and ability.

The proposed Local Tuition Discovery Platform, TutorLinkAI, is aimed at addressing the various limitations of the traditional methods of tutor discovery. By aggregating tutor related information and offering a wide

range of features for both parents and tutors, TutorLinkAI facilitates the tutor discovery process based on a number of criteria including subject, location, qualifications, availability, experience and affordability. By providing tutors with the ability to post their information and experience for students and parents to view, the proposed solution addresses the limitations of relying on word of mouth by offering both parents and students a wider pool of tutors to consider. Additionally, the proposed platform enables students and parents to sort through the available tutors according to their specific needs, thus eliminating the need to manually sift through hundreds of tutors or advertisements.

The proposed Local Tuition Discovery Platform, TutorLinkAI, supports the United Nations Sustainable Development Goal (SDG) 4: Quality Education by enabling students to receive more personalized supplementary education. By enabling proper matching of students and tutors based on student requirements as well as tutor qualifications, which focus on enhancing personalized learning and promoting access to quality education for all.

Apart from supporting universal access to quality education, TutorLinkAI aligns with SDG 10: Reduced Inequalities, which emphasizes the need for empowering economically disadvantaged individuals. By providing affordable means of accessing supplementary education and offering students and parents the option of sorting tutors based on their budget, TutorLinkAI can significantly help reduce the disparities associated with access to additional academic services.

As such, the proposed Local Tuition Discovery Platform, TutorLinkAI, is aimed at enabling personalized education by addressing the inefficiencies associated with accessing information regarding local tuition services. TutorLinkAI makes it easier and more transparent for you to find potential tutors. It combines local tuition services in one place and has more information on each tutor's qualifications and experience, as well as filtering and matching tools to match tutors with students based on their needs and budgets.

II. LITERATURE SURVEY

The development of an intelligent tutor recommendation system required consideration of both educational personalization and tutor reliability. Tutor recommendation involves more than matching a learner with a tutor who teaches the subject. The student's specified requirement from the tutor, the methodology of requirement representation and the reliability of the tutor's profile can all affect the metrics involved for recommendation.

Previous research has examined these aspects through educational recommender systems, learner modelling, semantic representation and credential and identity verification.

A. Educational Recommender Systems

Educational recommender systems have been developed to provide personalized recommendations based on the student's characteristics and their learning needs. Da Silva *et al.* [8] presents a systematic literature review of educational recommender systems and identifies personalization as one of the major objectives. It discusses the use of learner characteristics, contextual information and recommendation techniques in generating educational recommendations.

Achon *et al.* [2] proposed a recommendation system that considers tutor competencies and personality-related characteristics, showing that tutor selection can involve more than subject compatibility. Bhatt *et al.* [6] examined teacher-centric educational recommender systems and discussed factors related to recommendation accuracy and utility in educational settings. These studies demonstrate the importance of personalization, but a general learner profile may not fully represent the reason for seeking tutoring at a particular time. A student may require assistance with a specific topic while having different needs and constraints for other subjects. This creates the need for depiction of user needs separately than a generalized

B. Semantic Representation for Tutor Recommendation

Keyword based matching can be restrictive when students and tutors use different terms to describe related topics. Sentence embedding systems provide an alternative by representing text as numerical vectors and measuring semantic similarity.

Ni *et al.* [10] introduced a sentence encoding approach called Sentence - T5 which is based on pretrained text-to-text models, and investigated its application to semantic textual similarity. Wang *et al.* [13] investigated sentence similarity computation through joint FrameNet and element-focused Sentence-BERT framework models. These approaches are relevant to tutor recommendation because a student's learning requirement and a tutor's expertise may have similar meanings without sharing the same keywords. Semantic similarity helps identify relationships of the wordings listed. Although it should not be the only criterion. Therefore our system combines semantic similarity with explicit topic compatibility and structured information about the student and the tutor.

C. Learner Modelling and Learning Needs

Learner modelling provides methods for representing information about individual learners. Abdelrahman *et al* [1] surveyed knowledge tracing approaches for modelling students' evolving knowledge states, while Khosla *et al.* [9] investigated knowledge tracing based student skill assessment. Although these approaches focus on knowledge acquisition rather than tutor recommendation, they demonstrate the importance of learner-specific information in adaptive educational systems. TutorLinkAI focuses on applying this idea through a multiple-learning need representation. A student can maintain separate learning needs such as subjects, learning needs, academic level, preferred learning mode, language preference and budget. Recommendations can hence be generated on the basis of the chosen metrics rather than relying on a general student profile.

D. Credential and Identify Verification

Tutor recommendation systems raise questions related to the credibility of the tutor along with doubts on the information provided about the tutor and whether they can be trusted. Al-Ameri *et al.* [3] investigated automated document forgery detection, while Al Hemaury *et al.* [4] proposed a blockchain based framework for validation and authentication of academic certificates.

Identity verification has also been explored using facial information. Ali *et al* [5] investigated secure facial verification templates using deep learning and random projections. These studies provide us with a base on how verification layers can be implemented into systems. TutorLinkAI incorporates credential processing and identity evidence, including facial recognition and verification where applicable, as a part of tutor verification workflow before verified tutors become eligible for recommendation.

E. Research Gap

Although existing studies have addressed personalized educational recommendation, semantic similarity, learner modelling, and credential verification, these areas are commonly treated as separate components. Educational recommender systems primarily focus on identifying suitable learning support based on learner or tutor characteristics [2], [6]–[8], while semantic representation methods improve the comparison of textual requirements and expertise [10], [13]. Credential and identity verification research, in contrast, focuses on establishing the reliability of academic documents and personal identity [3]–[5], [11], [12].

A limitation of treating these aspects independently is that recommendation quality and tutor reliability are not considered together at the point of tutor selection. A tutor

may have strong subject compatibility but insufficient verification evidence, while a verified tutor may not be suitable for the learner's particular requirement. In addition, general student profiles may not adequately represent the specific learning objective for which tutoring is required.

To address these limitations, TutorLinkAI combines learning-need representation, tutor compatibility, and tutor verification within a single workflow. Individual learning needs are used to represent the student's current requirements, while semantic similarity, topic compatibility, and contextual constraints are used to rank suitable tutors. Tutor credentials and identity evidence are processed through a separate verification stage before tutors become eligible for recommendation. The system further provides explanations for recommendation outcomes and supports counterfactual analysis to examine how changes in learner constraints affect the resulting recommendations.

This establishes an integrated approach to tutor recommendation in which the suitability of a tutor and the reliability of the tutor profile are considered together rather than as independent processes.

III. EXISTING SYSTEM AND PROBLEM IDENTIFICATION

Existing solutions are mostly old-fashioned coaching institutes where their main goal is to increase student enrollment every year. This causes a financial drain on families and prevents students from finding the right help. These coaching institutes often take classes for 20-30 students at once, leading to a lack of personalisation and failing to cater to the individual needs of each student.

Most centers also assign the students homework assignments, adding to the physical and mental exhaustion of the students, which in turn again leads to them not performing as expected academically.

Although one can argue the demerits of the existing traditional methods, it is also important that we recognise the few centers that actually upskill the students and help them improve their academic capabilities through their extensive coaching.

Some existing online tutor finder platforms map the students to tutors who reside in a different pincode; this adds to the physical exhaustion of the students.

The platforms often advertise their fancy courses, which usually end up as clickbait for increased enrollments;

additionally, students having to choose from a wide variety of courses and faculties can become hectic.

Students usually have limited information about the availability of tutors in their localities, tutors who can help them in the subjects they are struggling with, and help them find tutors who charge fees that are economical for them.

Major Drawbacks of Existing Solutions

There are several ways for students to find tutors, but most of the existing methods still have some drawbacks.

1. Commercial Coaching Institutes and Manual Tuition Centers

In most cases, students find local tuition centers or coaching classes through advertisements, recommendations, or by asking around. Often the classes can end up being expensive, or they can be taught in groups, hence leaving the tutor to not be able to cater to individual student needs.

Finding the right tutor can also take time because parents often have to make calls or visit different centres.

2. Web and Directory-Based Tutor Portals

Examples: A+ Home Tuition, bdtutors.com and similar websites.

Online websites let students search for tutors using basic details like subject and location, and do not focus on important factors like student needs, affordability, and class scheduling. These platforms do not have direct communication between the tutors and the parents, and hence arranging classes may take extra time. The matching is also mostly based on profile details rather than what the student actually needs help with.

3. Multi-Criteria and Location-Based Platforms

Examples: GRAB Tutor Management System (GTMS), Tutors Map and Tutor Finder (LBS).

These platforms allow students to search for tutors using filters based on subject, gender, and expertise. They mainly use the information given in the tutor's profile. These platforms do not consider factors such as students' weak areas, teaching quality, availability, and fees.

Overall, most existing solutions focus on finding tutors using basic information or past activity. They do not give enough importance to the student's actual learning difficulties, current performance, tutor availability, and cost.

IV. PROPOSED METHODOLOGY

The system TutorLinkAI is a secure and intelligent platform that is designed to help parents and students identify the right tutors who are suitable for the students and are credible. The application addresses two major limitations that are existing in the current existing systems focusing on the tutor selection system, i.e., parents and students often find it difficult to assess the tutor suitability and the lack of reliable tutor verification in the existing system.

The proposed system consists of a tutor onboarding module, credential verification module, tutor profile management module, and tutor discovery and matching module. During onboarding, tutors register on the platform by providing their personal, academic, and professional information. An authentication mechanism is used to verify the tutor's account before the profile is created.

The key component of the system TutorLinkAI is the tutor credential verification process. In the application, the tutors are required to upload relevant documents that prove their qualification as a part of the onboarding process. The uploaded documents are then processed using Optical Character Recognition (OCR) to extract the relevant textual information. This extracted information is then analysed and compared with the information provided by the tutor during the registration process. Similarity-based validation is used to identify the inconsistencies between the submitted details and the registered profile information. Based on the verification results, the system will be able to identify whether the submitted information is consistent with the registered tutor profile and assigns an appropriate verification status.

The backend of the application is implemented using Python and FastAPI, this provides RESTful APIs for authentication, onboarding, profile management, document processing and verification. PostgreSQL is used for persistent data storage, with SQLAlchemy providing database interaction and Alembic supporting database schema migration. The frontend provides interfaces for tutor registration, profile creation, credential submission, and tutor discovery.

In the application, once the tutors are verified, the students and parents can view their profiles and identify tutors based on relevant academic requirements. The application combines identity onboarding, credential verification and tutor suitability assessment within one platform, hence providing a structured alternative to relying solely on recommendations from relatives, neighbors or nearby tutors.

V. SYSTEM DESIGN AND IMPLEMENTATION

The proposed TutorLinkAI system is designed as a modular intelligent platform for tutor discovery and personalized tutor recommendation. The system integrates tutor onboarding, learning need management, tutor verification, AI based recommendation, booking, session management and feedback. The implementation separates the frontend, backend and AI based matching components, verification system and database so that each part is constructed and evaluated individually. The workflow begins with student and tutor registration and profile creation. Students can define individual lesson needs, while tutors provide their areas of expertise and methods of tutoring along with these the tutors provide their availability, pricing, and supporting credentials. The verification flow processes the present tutor credentials and identity evidence, after which the eligible tutor profiles can participate in the recommendation process. For a selective learning need the matching component evaluates the tutor compatibility using semantic and topic based information together with contextual constraints. The resulting recommendation consists of the factors supporting the recommendation and the students get an overview of the reason for recommendation basis which they can select the tutor they wish to proceed with, post session the student is provided with a feedback page through which the feedback based evaluation of the tutor is completed for recommendation.

A. System Architecture

TutorLinkAI follows a layered client-server architecture consisting of a React-based frontend, a FastAPI backend, AI based matching services, verification services and PostgreSQL database. There are role based frontend interfaces present for the main two roles the student and the tutor, it handles operations such as registration, profile management, learning-need management, tutor discovery, booking, sessions and reviews. The backend service handles authentication, profiles, learning need management, recommendations, booking sessions and review of sessions. AI services perform semantic embedding and tutor ranking, while verification services process certificates and identity evidence.

TutorLinkAI — Layered System Architecture

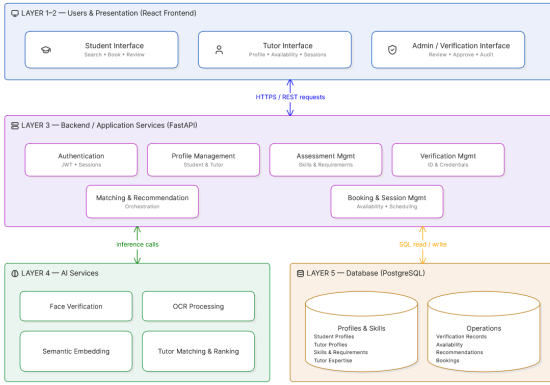


Fig. 1. Overall system architecture of TutorLinkAI.

B. Student and Tutor Profiling

Separate onboarding workflows are provided for students and tutors. Student profiles contain academic and preference information, while tutor profiles have subjects of expertise, availability, teaching preferences, pricing and qualification details. The collected information then forms the basis for learning need identification, verification, matching and recommendation processes.

1) Student Requirement Profile

The student's learning requirements are collected during onboarding and represented through individual learning needs. Instead of relying only on selected subjects, the system captures specific topics of interest, learning goals, academic level, preferred learning mode, language preferences, and budget. Multiple learning needs can be maintained by the same student, allowing matching components to use the requirement relevant to the current tutoring request.

2) Tutor Expertise Profile

Tutor expertise is represented using information such as subjects, topics, skills, availability, teaching preferences, pricing, qualifications and teaching expertise. Both structured attributes and descriptive expertise information can be utilized during the matching process.

3) Profile and Requirement Management

Student and tutor information is maintained separately from the recommendation results. Students can update their learning needs, while tutors can update relevant profile and expertise information. The resulting profile and requirement data are used by the verification and recommendation components when processing a tutor discovery request.

C. Learning-Need Representation

TutorLinkAI separates a student's tutoring requirements from the general profile and allows multiple learning needs to be maintained. Each learning need contains information such as subject, topics, learning goals, academic level, learning mode, language preference and budget. A selected learning need is used as a primary input to the recommendation process, allowing recommendations to reflect the student's current objective.

D. Tutor Verification

Tutor verification is incorporated to improve the reliability and credibility of tutor profiles. The verification module combines facial verification and document-based verification.

1) Face Verification

The system processes the tutor's submitted facial image to obtain a facial representation for identity comparison. OpenCV can be used for image processing, with InsightFace which is used for feature detection and facial structure detection. The resulting similarity information is used as one of the signals in determining the tutor's verification status [5], [12].

2) Certificate Verification

Uploaded qualification certificates are processed using Optical Character Recognition (OCR). PaddleOCR is used to extract relevant information from certificates such as the tutor's name, institution, qualification and other available textual fields. OCR-based extraction provides structured information that is compared against the tutor's profile information with stronger institutional validation as a future extension [4], [11].

E. AI Based Tutor Matching

The matching component compares a selected learning need with eligible tutors using systems like semantic similarity, topic compatibility, and contextual constraints. Student requirements and tutoring expertise are encoded using all-MiniLM-L6-v2, producing 384-dimensional embeddings. Cosine similarity is used for semantic comparison.

When topic information is available:

$$S_{\text{learning}} = 0.60 S_{\text{semantic}} + 0.40 S_{\text{topic}}$$

The overall recommendation score is evaluated as:

$$S_{\text{overall}} = 0.45 S_{\text{learning}} + 0.20 S_{\text{location}} + 0.20 S_{\text{fee}} + 0.15 S_{\text{time}}$$

This combines the learning compatibility with location, budget and availability

F. Explainable and Counterfactual Recommendation

Recommendations include supporting factors such as matched topics, semantic alignment, academic level, learning mode, language, budget and schedule compatibility. The system also supports counterfactual analysis, allowing recommendation rankings to be examined after changing constraints such as the student's budget.

G. Booking, Session Management and Feedback

Selected tutors can be booked through the platform, with confirmed bookings associated with session records. Sessions support the states as Schedules, In Progress, Completed and Cancelled. After the completion of a session students can submit feedback on the session.

An experimental feedback component applies Bayesian smoothing to ratings to reduce the influence of small review counts. This component is used for research analysis and does not replace the primary production ranking.

H. Research Evaluation Framework

The implementation system was evaluated through multiple experiments covering recommendation quality, component contribution, feedback sensitivity, and tutor verification. Different recommendation configurations were compared using Precision @1/3/5, Recall@5, MRR, NDCG @%, and pairwise ranking accuracy. Ablation experiments were conducted to examine the contribution of semantic and topic-wise matching components. Metrics such as feedback-weight and cold-start experiments were used to study the effect of tutor ratings on recommendation ranking. The tutor verification pipeline was evaluated using certificate processing, profile consistency, facial evidence, liveliness and identity similarity outcomes. These experiments provide a controlled and reproducible evaluation of the implemented components, while broader real world generalisation would require evaluation using larger real world datasets.

VI. RESULTS AND DISCUSSIONS

The implementation of TutorLinkAI demonstrates that tutor onboarding and credential verification can be combined within a single tutoring platform. The system guides tutors step by step through signing up, submitting their qualifications, and being checked before they are listed as prospective tutoring resources.

Early in development, we found that registration and verification cannot be handled at the same time: a tutor account does not prove that the degree behind it is genuine. Separating the two stages gives the platform a real basis for

labeling a tutor "verified" rather than merely "signed up." This is a major change in how parents make decisions. Instead of choosing a tutor because a neighbor mentioned a name or because someone lives two streets away, parents can weigh qualifications and verification status alongside location, which is a very different and arguably more responsible way to decide who teaches their child.

The approach has limitations, however. OCR is only as good as the document it receives, and a blurry photograph, poor handwriting, or an unusual certificate format can reduce extraction accuracy. The similarity check has a comparable ceiling: it can detect that the extracted text does not match the registration form, but it cannot confirm that the issuing institution is legitimate unless an external database is available to check against. For now, therefore, the verification layer serves as an automated first pass at assessing credibility, not a replacement for an actual institutional check.

VII. CONCLUSION

TutorLinkAI is tackling a problem which has for a long time been dealt with inadequately: how can a parent be sure that the tutor they are about to hire actually knows what they say they know? Instead of depending on word of mouth, the platform uses a formal onboarding process together with automated checks of credentials and provides tutors with a real profile to back up their claims. Parents now get something that has never been available to them before: a tutor profile complete with an actual verification status, rather than just a name and a phone number given to them by a relative.

This is the aim that TutorLinkAI has set for itself, namely to eliminate the guesswork involved in selecting a tutor and replace it with an assessment that a parent can make. For tutors, the platform provides a real benefit: one who has qualifications now has a clear method of proving them, rather than having to rely on a resume which no one ever checks. Over time, this will help to distinguish qualified educators from those who are only good at self-promotion.

This version shows that it is possible to incorporate credential verification into a tutoring platform and for it to function without the need for manual examination of each document, so that parents can rely on concrete information rather than on a suggestion from someone who lives two streets away. That is the gap which TutorLinkAI intended to close, and in this respect it achieves what it set out to do.

VIII. FUTURE SCOPE

The project TutorLinkAI, provides a proper foundation for automated verification and recommendation of tutors and

mentors. Furthermore we can build upon this existing model by improving credential verification, student performance analysis, security and personalized learning support.

1. Institutional Credential Verification:

The existing system TutorLinkAI can be integrated with the universities, educational boards and certification databases to verify the actual credibility of the tutors from authorised tutors, making the entire verification process trustworthy and seamless. This mechanism can improve the reliability of the verification process and reduce invalid credentials submissions[4], [11].

2. Advanced Document Fraud Detection:

We can also incorporate image-forensics and machine-learning techniques to identify duplicates, forged, malicious, or digitally altered documents by providing an additional layer of security in the tutor verification process.[3].

3. Early Student Performance Prediction:

Incorporating machine learning models for predicting the student performance at early stages can help generate a personalized learning pattern. This can be done by analysing the student learning patterns, assessment results and the interaction data to identify the students who may be at risk for poor performance. With the help of these early detection systems, we can provide the students with timely feedback and targeted support[1], [9].

4. Feedback and Rating Analysis:

Using Natural Language Processing (NLP) techniques, we can analyse the student and parent

feedback to tutoring sessions and identify common patterns in tutor performance. These insights support the students in making informed decisions while choosing the right tutor for them.

5. Continuous Tutor Verification:

The system can periodically reassess and reverify the tutor credentials and profile information to ensure that all the verification remains current.

6. Scalable Cloud Deployment:

Cloud infrastructure and containerization can be adopted to support an increasing number of students, tutors and concurrent verification requests.

7. Enhanced Security and Privacy:

Future implementations can incorporate stronger encryption, role-based access control, audit logging, and privacy preserving storage mechanisms to protect sensitive credentials and academic documents[5], [12].

8. Multimodal AI-Based Doubt Solving:

The platform can be extended with a multimodal AI module which is capable of processing text, images and other resources for providing the students with learning aid and resolving academic doubts. This system can provide interactive and context-aware learning support alongside tutor recommendations.

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